

Climbing the Leaderboard

An arcade game player wants to climb to the top of the leaderboard and track their ranking. The game uses [Dense Ranking](#), so its leaderboard works like this:

- The player with the highest score is ranked number **1** on the leaderboard.
- Players who have equal scores receive the same ranking number, and the next player(s) receive the immediately following ranking number.

Example

ranked = [100, 90, 90, 80]

player = [70, 80, 105]

The ranked players will have ranks **1**, **2**, **2**, and **3**, respectively. If the player's scores are **70**, **80** and **105**, their rankings after each game are **4th**, **3rd** and **1st**. Return [4, 3, 1].

Function Description

Complete the *climbingLeaderboard* function in the editor below.

climbingLeaderboard has the following parameter(s):

- *int ranked[n]*: the leaderboard scores
- *int player[m]*: the player's scores

Returns

- *int[m]*: the player's rank after each new score

Input Format

The first line contains an integer *n*, the number of players on the leaderboard.

The next line contains *n* space-separated integers *ranked[i]*, the leaderboard scores in decreasing order.

The next line contains an integer, *m*, the number games the player plays.

The last line contains *m* space-separated integers *player[j]*, the game scores.

Constraints

- $1 \leq n \leq 2 \times 10^5$
- $1 \leq m \leq 2 \times 10^5$
- $0 \leq ranked[i] \leq 10^9$ for $0 \leq i < n$
- $0 \leq player[j] \leq 10^9$ for $0 \leq j < m$
- The existing leaderboard, *ranked*, is in *descending* order.

score max 10^9

always? 10^9 10^9

- The player's scores, *player*, are in ascending order.

→ jđ ga bakal naik-turun-naik

Subtask

For 60% of the maximum score:

$$1 \leq n \leq 200$$

$$1 \leq m \leq 200$$

Sample Input 0

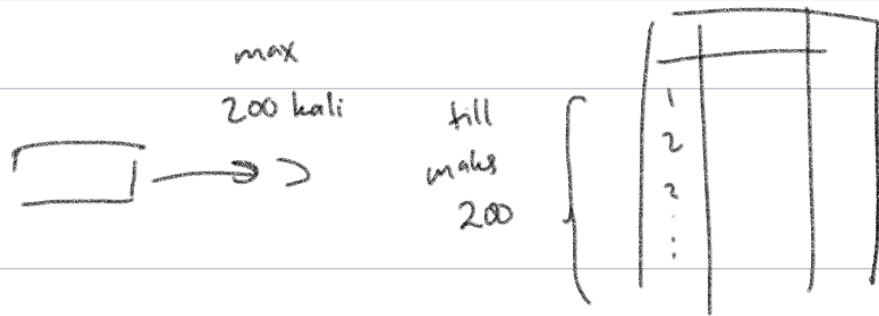
```
7
100 100 50 40 40 20 10
4
5 25 50 120
```

n = banyak player di leaderboard

m = banyak game yg dimainkan player ltr

Sample Output 0

```
6
4
2
1
```



Explanation 0

Alice starts playing with 7 players already on the leaderboard, which looks like this:

Rank	Name	Score
1	Emma	100
1	David	100
2	Caroline	50
3	Ritika	40
3	Tom	40
4	Heraldo	20
5	Riley	10

After Alice finishes game 0, her score is 5 and her ranking is 6:

Rank	Name	Score
1	Emma	100
1	David	100
2	Caroline	50
3	Ritika	40
3	Tom	40
4	Heraldo	20
5	Riley	10
6	Alice	5

After Alice finishes game 1, her score is 25 and her ranking is 4:

Rank	Name	Score
1	Emma	100
1	David	100
2	Caroline	50
3	Ritika	40
3	Tom	40
4	Alice	25
5	Heraldo	20
6	Riley	10

After Alice finishes game **2**, her score is **50** and her ranking is tied with Caroline at **2**:

Rank	Name	Score
1	Emma	100
1	David	100
2	Caroline	50
2	Alice	50
3	Ritika	40
3	Tom	40
4	Heraldo	20
5	Riley	10

ranked[j]
 $\text{player}[i] == \text{ranked}[j-1] \text{ and } \text{player}[i] >$
 $\text{player_rank.append(leaderboard_rank[j-1])}$

After Alice finishes game **3**, her score is **120** and her ranking is **1**:

Rank	Name	Score
1	Alice	120
2	Emma	100
2	David	100
3	Caroline	50
4	Ritika	40
4	Tom	40
5	Heraldo	20
6	Riley	10

Sample Input 1

```
6
100 90 90 80 75 60
5
50 65 77 90 102
```

Sample Output 1

```
6
5
4
```

leaderboard_rank = n * [1]

for i in range(n-1):

if ranked[i+1] == ranked[i]:

leaderboard_rank[i+1] = leaderboard_rank[i]

elif ranked[i+1] < ranked[i]:

leaderboard_rank[i+1] = leaderboard_rank[i] + 1

di emi

Ok ajr

belly- y

perken

time limit (?)

~~for i in range(m):~~

~~for j in range(n, -1, -1):~~ → mungkin bisa dipersempit di emi.

~~if player[i] <= ranked[j]:~~ Ga perlu loop semu ya ada di ranked,

~~player_rank.append(leaderboard_rank[j] + 1)~~ tapi cukup di selitiran
~~break~~ skor player kita aja.

~~elif player[i] > ranked[j] and player[i] < ranked[j-1]:~~

~~if leaderboard_rank[j] > 1:~~

~~player_rank.append(leaderboard_rank[j] + 1)~~
~~break~~

1 70
60
2 50

~~elif leaderboard_rank[j] == 1:~~

~~elif player[i] >= ranked[j-1] and player[i] > ranked[j] and~~
~~leaderboard_rank[j] == 1:~~

~~player_rank.append(1)~~

~~break~~

j = 5
j = 4

80
1 70
2 50

~~time limit 1st ATTEMPT~~

range terdekat dari skor player[i] ?

apa da nilai jarak terkecil antara skor player[i] dan skor org di leaderboard

ranked = [...]

nearest_score = min(ranked, key = lambda x: abs(x - player[i]))

if nearest_score < player[i]:

~~49~~ ~~48~~ 80
~~50~~ 49 70

current_rank = leaderboard_rank[ranked.index(nearest_score)]

player_rank.append(current_rank) 45 90

elif nearest_score > player[i]:

46 80

current_rank = leaderboard_rank[ranked.index(nearest_score)] + 1

player_rank.append(current_rank)

else:

current_rank = leaderboard_rank[ranked.index(nearest_score)]

player_rank.append(current_rank)

Ok. let's try this attempt DD:

25 ? ~~→~~ 4 ✓

ah. as always - debug time * _ *

4 20

ternyata cara ini mangle henc time limit.

2 50

1 120

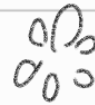
harus pangle lari.

2 50

2 + 1 100

~~X~~ Time Limit 2nd ATTEMPT

namanya GPT tehehe - mentok :3



ada yang namanya bisect.

contoh implementasinya:

```
from bisect import bisect_left
```

data = [3, 8, 15, 20, 35]

```
pos = bisect_left(data, target)
```

target = 18

```
print(pos)
```

>> output: 3

* reverse [2, 1, 0] → [0, 1, 2]

ranked = ranked[::-1]

leaderboard_rank = len_n[::-1]

artinya, target = 18 dalam arraynya ke 1st data sampai ttp terunt, masukinya ke index 3.

jd gini → data = [3, 8, 15, 18, 20, 35]

rank	Score
5	3 0
4	8 1
3	15 2
→ 2	18 rank 3
2	20 3
1	35 4

if player[i] > ranked[pos-1] and player[i] < ranked[pos]:

player_rank.append(leaderboard_rank[pos-1])

~~elif player[i] == ranked[pos]:~~ else:

append [pos]

~~elif~~

len(ranked) = 5

*/ pos: 5 dan player[i] = 35 ? → rank 1

> 35? → rank 1